

Databricks Generative AI

Questions 2-29 | All options and correct answers

Question 2

A Generative AI Engineer is building an LLM to generate article summaries in the form of a type of poem, such as a haiku, given the article content. However, the initial output from the LLM does not match the desired tone or style. Which approach will NOT improve the LLM's response to achieve the desired response?

- A. Include few-shot examples in the prompt to the LLM.
- B. Fine-tune the LLM on a dataset of the desired tone and style.
- C. Provide the LLM with a prompt that explicitly instructs it to generate text in the desired tone and style.
- D. Use a neutralizer to normalize the tone and style of the underlying documents.

Correct answer: D. Use a neutralizer to normalize the tone and style of the underlying documents.

Question 3

A Generative AI Engineer would like an LLM to generate formatted JSON from emails. This requires parsing and extracting the order ID, date, and sender email. A sample email contains Date: April 23, 2024; From: anjali.thayer@computex.org; and the message: 'I have a shipment (order ID is CD34RFT). Can you please send me an update?' Which prompt will extract the relevant information in JSON format with the highest level of output accuracy?

- A. You will receive customer emails and need to extract date, sender email, and order ID.
- B. You will receive customer emails and need to extract date, sender email, and order ID. Return the extracted information in JSON format.
- C. You will receive customer emails and need to extract date, sender email, and order ID. Return the extracted information in JSON format. Here is an example: {"date": "April 16, 2024", "sender_email": "sarah.lee925@gmail.com", "order_id": "RE9987D"}.
- D. You will receive customer emails and need to extract date, sender email, and order ID. Return the extracted information in a human-readable format.

Correct answer: C. You will receive customer emails and need to extract date, sender email, and order ID. Return the extracted information in JSON format. Here is an example: {"date": "April 16, 2024", "sender_email": "sarah.lee925@gmail.com", "order_id": "RE9987D"}.

Question 4

A Generative AI Engineer is evaluating a customer-support agent in Databricks. The team needs to score each response on a domain-specific policy: the answer must cite an approved refund rule and must not mention unsupported escalation paths. Built-in evaluation metrics do not capture this logic. The team wants the metric to run during agent evaluation in Databricks and return a repeatable structured score for each trace. Which approach should the engineer use?

- A. Create a custom MLflow scorer that inspects agent outputs against the policy and pass it into the evaluation run.
- B. Use only latency and token-count metrics because custom policy checks are not supported in evaluation workflows.
- C. Log the traces to MLflow 3.x and review failures in the UI without defining a scorer.
- D. Add the policy text to the system prompt and rely on the model's self-reported compliance as the evaluation result.

Correct answer: A. Create a custom MLflow scorer that inspects agent outputs against the policy and pass it into the evaluation run.

Question 5

A Generative AI Engineer has set up an endpoint with AI Guardrails turned on to block incoming requests that divulge PII. Inference tables are enabled for this endpoint. If an end user sends their phone number in the prompt, what will appear in the inference table for that record?

- A. The request will appear, but the response will be the error indicating the request was blocked due to PII.
- B. The full answer that would have been sent without the PII block will appear in inference tables.
- C. The record will not appear in inference tables because it contains PII.
- D. The request will appear, but the response column will be blank.

Correct answer: A. The request will appear, but the response will be the error indicating the request was blocked due to PII.

Question 6

A Generative AI Engineer is deploying an agent using Mosaic AI Model Serving. The agent needs to access Databricks resources including Vector Search, Databricks SQL, and Functions. What is the easiest, best-practice way to authenticate the deployed agent to access these resources?

- A. Define resource dependencies while logging the agent and deploy it with the Agent Framework.
- B. Set appropriate permissions on the Model Serving endpoint for the agent, as these permissions will be used when connecting to other resources.
- C. Log the authentication token while logging the agent; this token will be automatically used for authentication.
- D. Embed authentication credentials within the agent's code to access the required resources.

Correct answer: A. Define resource dependencies while logging the agent and deploy it with the Agent Framework.

Question 7

A Generative AI Engineer is integrating Mosaic AI Vector Search into a Retrieval-Augmented Generation (RAG) system. The source data, comprising simple text entries, is stored in a Delta table. To simplify the workflow, the engineer plans to use an embedding model served via a Mosaic AI Model Serving endpoint to automatically compute embeddings during data synchronization from the Delta table to the vector search index. Which method should the engineer use?

- A. Direct vector access index.
- B. Delta Sync index with self-managed embeddings.
- C. Hybrid embedding computing.
- D. Delta Sync index with managed embeddings.

Correct answer: D. Delta Sync index with managed embeddings.

Question 8

A Generative AI Engineer has been asked to design an LLM-based application that answers employee HR questions using HR PDF documentation. Which set of high-level tasks should the system perform?

- A. Split HR documentation into chunks and embed them into a vector store. Use the employee question to retrieve the best-matched chunks and use the LLM to generate a response based on the retrieved documentation.
- B. Create an interaction matrix of historical employee questions and HR documentation. Use ALS to factorize the matrix, create embeddings for new queries, find the best documentation, and generate a response.
- C. Use an LLM to summarize all HR documentation. Provide the summaries and user query to an LLM with a large context window to generate a response.
- D. Calculate averaged embeddings for each HR document, compare them with the user query to find the best document, and pass the entire document with the query to an LLM.

Correct answer: A. Split HR documentation into chunks and embed them into a vector store. Use the employee question to retrieve the best-matched chunks and use the LLM to generate a response based on the retrieved documentation.

Question 9

A Generative AI Engineer has already trained an LLM on Databricks, and it is now ready to be deployed. Which steps correctly outline the easiest process for deploying a model on Databricks?

- A. Log the model as a pickle object, upload it to a Unity Catalog volume, register it to Unity Catalog using MLflow, and start a serving endpoint.
- B. Log the model using MLflow during training, register the model directly to Unity Catalog using the MLflow API, and start a serving endpoint.
- C. Wrap the LLM's prediction function in a Flask application and serve it using Gunicorn.
- D. Save the model and dependencies in a local directory, build a Docker image, and run the Docker container.

Correct answer: B. Log the model using MLflow during training, register the model directly to Unity Catalog using the MLflow API, and start a serving endpoint.

Question 10

A Generative AI Engineer at an electronics company deployed a RAG application for customers to ask questions about products the company carries. Feedback indicates that responses often contain information about an irrelevant product. What can the engineer do to improve response relevance?

- A. Assess the quality of the retrieved context.
- B. Use a different LLM to improve the generated response.
- C. Use a different semantic similarity search algorithm.
- D. Implement caching for frequently asked questions.

Correct answer: A. Assess the quality of the retrieved context.

Question 11

A Generative AI Engineer has a provisioned-throughput Model Serving endpoint as part of a RAG application and wants to monitor incoming requests and outgoing responses. The current approach uses a microservice between the endpoint and the user interface to write logs to a remote server. Which Databricks feature should be used instead to perform the same task?

- A. Lakeview.
- B. DBSQL.
- C. Inference Tables.
- D. Vector Search.

Correct answer: C. Inference Tables.

Question 12

A Generative AI Engineer is creating an LLM-powered application that needs access to up-to-date news articles and stock prices. Stock prices are stored in Delta tables, and the latest relevant news must be found by searching the internet. How should the engineer architect the system?

- A. Create an agent with tools for SQL querying of Delta tables and web searching, then provide retrieved values to an LLM to generate the response.
- B. Use an LLM to summarize the latest news articles and look up stock tickers from the summaries to find stock prices.
- C. Download and store news articles and stock-price information in a vector store. Use a RAG architecture to retrieve and generate at runtime.
- D. Query the Delta table for volatile stock prices and use an LLM to generate a search query to investigate potential causes of the volatility.

Correct answer: A. Create an agent with tools for SQL querying of Delta tables and web searching, then provide retrieved values to an LLM to generate the response.

Question 13

A Generative AI Engineer is developing an application based on an open-source large language model. They need a foundation LLM with a large context window. Which model fits this need?

- A. MPT-30B.
- B. DBRX.
- C. Llama2-70B.
- D. DistilBERT.

Correct answer: B. DBRX.

Question 14

A Generative AI Engineer is building a RAG application that answers questions about internal documents for SnoPen AI. Source documents may contain significant irrelevant content, such as advertisements, sports news, entertainment news, or content about other companies. Which approach is advisable for filtering irrelevant information?

- A. Include in the system prompt that the application must not answer questions unrelated to SnoPen AI.
- B. Consolidate all SnoPen AI-related documents into a single chunk in the vector database.
- C. Keep all articles because the RAG application needs to understand non-company content to avoid answering questions about it.
- D. State in the system prompt that any information it sees is about SnoPen AI, even if no data filtering is performed.

Correct answer: A. Include in the system prompt that the application must not answer questions unrelated to SnoPen AI.

Question 15

A team uses Mosaic AI Vector Search to retrieve documents for a RAG pipeline. A search returns five documents, and the first three are added to the prompt. Agent Evaluation shows that some lower-ranked documents have higher context-relevancy scores than higher-ranked documents. Which option should the team consider?

- A. Use a reranker to order the documents based on relevance scores.
- B. Use a different embedding model for computing document embeddings.
- C. Modify the prompt to instruct the LLM to order the documents based on relevance scores.
- D. Increase the number of documents added to the prompt to improve context relevance.

Correct answer: A. Use a reranker to order the documents based on relevance scores.

Question 16

A Generative AI Engineer has created a RAG application for answering questions about a fantasy-novel series. The novel texts are chunked and embedded in a vector store with metadata. The engineer wants to choose chunking strategies and configurations methodically. Which TWO strategies should be used to optimize the chunking strategy and parameters?

- A. Add a classifier for user queries that predicts which book will best contain the answer, and use it to filter retrieval.
- B. Choose an evaluation metric such as recall or NDCG and experiment with chunking changes, such as splitting by paragraphs or chapters. Select the strategy with the best metric.
- C. Pass known questions and best answers to an LLM and ask it for the best token count. Use a summary statistic of the token counts to choose chunk size.
- D. Change embedding models and compare performance.
- E. Create an LLM-as-a-judge metric to evaluate how well previous questions are answered by the most appropriate chunk, and optimize chunking parameters using that metric.

Correct answers: B and E. Choose an evaluation metric such as recall or NDCG and experiment with chunking changes, such as splitting by paragraphs or chapters. Select the strategy with the best metric. / Create an LLM-as-a-judge metric to evaluate how well previous questions are answered by the most appropriate chunk, and optimize chunking parameters using that metric.

Question 17

A Generative AI Engineer is developing a RAG application for a small internal group of experts. They want the best possible answer quality; latency and throughput are not concerns. The topics and data are highly confidential, and regulatory requirements prohibit transmitting information to third parties. Which model meets all these needs?

- A. OpenAI GPT-4.
- B. BGE-large.
- C. Dolly 1.5B.
- D. Llama2-70B.

Correct answer: D. Llama2-70B.

Question 18

A Generative AI Engineer needs to allocate costs for an agent deployed with Agent Framework behind AI Gateway. Finance requires daily reporting by workspace and endpoint, including token usage and request counts. The engineer wants to query the data with SQL and join it to an internal cost-center mapping table. Which table type best serves this need?

- A. Inference Tables.
- B. System Serving, Served Entities table.
- C. MLflow experiment run tables.
- D. AI Gateway Usage table.

Correct answer: D. AI Gateway Usage table.

Question 19

A Generative AI Engineer is deploying an application that uses a custom MLflow PyFunc model to return interim results. How should the endpoint be configured to pass secrets and credentials?

- A. Use `spark.conf.set()`.
- B. Pass variables using the Databricks Feature Store API.
- C. Add credentials using environment variables.
- D. Pass the secrets in plain text.

Correct answer: C. Add credentials using environment variables.

Question 20

Databricks offers built-in AI judges that provide metrics and rationale for different quality issues. Which pair of judges both require a ground-truth label in the evaluation dataset field `expected_response` to execute?

- A. `relevance_to_query` and `chunk_relevance`.
- B. `correctness` and `groundedness`.
- C. `guideline_adherence` and `chunk_relevance`.
- D. `context_sufficiency` and `correctness`.

Correct answer: D. context_sufficiency and correctness.

Question 21

A Generative AI Engineer has deployed a customer-support agent that retrieves product documentation and generates responses. SMEs review responses through a web interface that captures ratings and written comments. The engineer needs to collect this feedback systematically and create an evaluation dataset for comparing future agent versions with the current baseline. Which approach should be used?

- A. Create a Unity Catalog view containing only 5-star interactions and use that view as the evaluation dataset for every future version.
- B. Export only the written comments to a text file, manually score them with a custom script, and use the script output as the evaluation dataset.
- C. Use the review app to collect feedback and directly deploy the highest-rated responses as the new baseline without creating a formal evaluation dataset.
- D. Log the SME ratings and comments to a Delta table with the corresponding user queries and agent responses, then use MLflow to create and register an evaluation dataset for future evaluations.

Correct answer: D. Log the SME ratings and comments to a Delta table with the corresponding user queries and agent responses, then use MLflow to create and register an evaluation dataset for future evaluations.

Question 22

A Generative AI Engineer is developing a patient-facing healthcare chatbot. If a patient's question is not a medical emergency, the chatbot should collect more information for the doctor's office and suggest pre-approved medical articles. If the question is urgent, it should direct the patient to local emergency services. Given the input, 'I have been experiencing severe headaches and dizziness for the past two days,' which response is most appropriate?

- A. Please provide your age, recent activities, and any other symptoms noticed with the headaches and dizziness.
- B. Please call your local emergency services.
- C. Here are a few relevant articles for your browsing. Let me know if you have questions after reading them.
- D. Headaches can be tough. Hope you feel better soon!

Correct answer: B. Please call your local emergency services.

Question 23

A Generative AI Engineer is building an interactive catalog for an inventory system that allows users to search for any item using a plain-text description. There are about 17,000 items, and new items are not frequently added. The solution must be cost-effective and easy to maintain. Which solution should the engineer choose?

- A. Storage-optimized vector search with a Direct Vector Access index and triggered sync.
- B. Standard vector search with Databricks-managed embeddings and a Delta Sync index with continuous sync.
- C. Standard vector search with self-managed embeddings and a Delta Sync index with continuous sync.
- D. Standard vector search with Databricks-managed embeddings and a Delta Sync index with triggered sync.

Correct answer: D. Standard vector search with Databricks-managed embeddings and a Delta Sync index with triggered sync.

Question 24

A Generative AI Engineer is building a compound AI system that processes customer event reports against a coding system and corporate guidelines. It must answer questions from guideline documents, extract event codes from reviewers' notes, and route ambiguous requests to the appropriate specialized handler. All capabilities must operate under one entry point that interprets intent and delegates. Which Agent Brick should be the top-level orchestrator?

- A. Knowledge Assistant, because the chatbot interface can handle multi-turn conversations.
- B. Knowledge Assistant, because it can be configured with multiple vector indexes to handle all three request types simultaneously.
- C. Multi-Agent Supervisor, because it interprets incoming requests and delegates tasks to specialized sub-agents.
- D. Multi-Agent Supervisor, because it can be used without Knowledge Assistant and Information Extraction agents.

Correct answer: C. Multi-Agent Supervisor, because it interprets incoming requests and delegates tasks to specialized sub-agents.

Question 25

A Generative AI Engineer is building a system that answers questions about the latest stock-news articles. Which option will NOT help ensure that outputs are relevant to financial news?

- A. Increase compute to improve question-processing speed and allow greater relevancy analysis.
- B. Incorporate manual reviews to correct problematic outputs before sending them to users.
- C. Implement a comprehensive guardrail framework with content-filter policies tailored to the finance sector.
- D. Implement a profanity filter to screen out offensive language.

Correct answer: D. Implement a profanity filter to screen out offensive language.

Question 26

Which statement accurately identifies the difference between the evaluation and monitoring phases in the generative AI application lifecycle within Databricks?

- A. Evaluation ensures responses comply with business rules in production, whereas monitoring focuses on SLA and performance metrics.
- B. Evaluation uses all inference history to assess production readiness, whereas monitoring uses only new inference-table records.
- C. Evaluation logs and traces live production API calls, whereas monitoring runs metrics on inference tables containing those traces.
- D. Evaluation uses Mosaic AI Agent Evaluation and an evaluation dataset to assess performance metrics and traces, whereas monitoring uses inference tables as source data for creating a metrics profile.

Correct answer: D. Evaluation uses Mosaic AI Agent Evaluation and an evaluation dataset to assess performance metrics and traces, whereas monitoring uses inference tables as source data for creating a metrics profile.

Question 27

A Generative AI Engineer is developing an LLM application that generates personalized birthday poems based on users' names. Which technique is most effective for safeguarding the application against potentially malicious user inputs?

- A. Increase the compute that powers the LLM so it processes input faster.
- B. Reduce the time users can interact with the LLM.
- C. Implement a safety filter that detects harmful inputs and instructs the LLM to respond that it is unable to assist.
- D. Ask the LLM to remind the user that the input is malicious but continue the conversation.

Correct answer: C. Implement a safety filter that detects harmful inputs and instructs the LLM to respond that it is unable to assist.

Question 28

A team wants to serve a code-generation model as an assistant for software developers. It should support multiple programming languages, and quality is the primary objective. Which Databricks Foundation Model API or Marketplace model is the best fit?

- A. CodeLlama-34B.
- B. MPT-7B.
- C. BGE-large.
- D. Llama2-70B.

Correct answer: A. CodeLlama-34B.

Question 29

What is the most suitable library for building a multi-step LLM-based workflow?

- A. PySpark.
- B. Pandas.
- C. TensorFlow.
- D. LangChain.

Correct answer: D. LangChain.